

GUARDIAN-AI

Smart ADAS and Real-Time Driver Safety Copilot

Hack2Skill Data & AI Challenge Submission | Built by Jatin

The Problem: Cognitive Distractions & Road Hazards

1.3 Million Annual Road Deaths: Global statistics show that the vast majority of severe vehicular accidents are caused by human error, primarily micro-sleeps or distractions.

- **Silent Fatigue (Micro-sleeps):** Drivers fall asleep for small intervals, leading to immediate lane drift and high-speed crashes.
- **Visual Distractions (Phone Usage):** Looking away to interact with devices blocks response times for obstacle avoidance.
- **Missing Immediate Closed-Loop Warnings:** Existing dashboard cameras only record feeds, lacking edge-computer vision models that calculate risk scores and warning signals locally in real-time.

Executive Summary: The GuardianAI Vision

GuardianAI bridges the critical gap between video capturing and physical accident prevention. By using lightweight edge computer vision in the browser and asynchronous multi-agent models in the backend, it acts as a proactive safety co-pilot.

- **Attentiveness Indexing:** Continuously scores driver focus from 0 to 100 based on blink frequency, head angles, and yaw deviation.
- **Asynchronous Telemetry:** Combines driver posture with live steering and braking loops to calculate stability.
- **Collision Forecasting:** Projects trajectory risk 3-5 seconds before impact, giving drivers enough time to take corrective action.

System Architecture: 4 Collaborating AI Agents

Our system implements an event-driven micro-agent architecture running asynchronously on a central event bus:

[Driver Behavior Agent]

Monitors face yaw, pitch, blink rate (EAR), and smartphone phone call postures.

[Prediction Agent]

Fuses driver and vehicle features to forecast accident probability and type.

[Vehicle Dynamics Agent]

Tracks speed, steering angle deviations, and road lane alignment values.

[Intervention Agent]

Escalates warnings (alarms, automatic deceleration instructions) to prevent crashes.

Client-Side Edge AI: MediaPipe Holistic CV

Rather than streaming heavy video feeds to the cloud, GuardianAI runs all computer vision models locally inside the driver's browser interface:

- **Yaw & Pitch angle estimation:** Uses scale-invariant nose-tip to eye ratios to catch left/right turns and up/down nodding.
- **Dual-Cheek proximity detection:** Checks wrist and finger landmarks to detect phone usage at either ear.
- **Zero Server GPU Cost:** Processing frames locally reduces backend workload and solves bandwidth constraints.

Backend Intelligence: PyTorch Neural Networks

All backend models are fully trained in PyTorch on generated sequence scenarios and saved as checkpoints:

Model Name	Architecture Type	File Checkpoint
Driver Monitor	MobileNetV2 CNN Classifier	driver_monitor.pt
Vehicle Dynamics	Bi-directional GRU Sequence	vehicle_behavior.pt
Sensor Fusion	Transformer Cross-Attention	sensor_fusion.pt
Accident Predictor	LSTM Sequence Forecaster	accident_predictor.pt

Safety Interventions Escalation Ladder

The Intervention Agent maps real-time risk scores directly to safety controls:

Risk Score	Severity	System Intervention Action
0 - 25%	LOW	Normal operations, background monitoring active.
25 - 50%	MEDIUM	Caution advisory tips played on screen, voice guidance starts.
50 - 75%	HIGH	Warning wailing sirens sound, lane alignment advisory active.
75 - 100%	CRITICAL	High-pitch wailing siren, automatic deceleration instruction sent.

Interactive Cockpit Controls & DOM Optimization

- **Telemetry Override sliders:** Control speed, steering angle, and lane alignment variables directly in the dashboard UI to simulate emergency risk conditions.
- **Direct DOM Node Updates:** Excludes high-frequency eye and landmark values from the React state cycle. Coordinates are updated directly via element IDs, preventing interface lag.
- **Throttled Sockets:** Transfers webcam parameters to the backend at 150ms intervals, avoiding websocket buffers overloading.

System Verification & Test Metrics

- **10/10 pytest Suite:** Verifies asynchronous event queues, database persistence logs, API routes, and network models successfully.
- **Inference Timing:** Neural network forward passes execute in under 10ms on CPU, leaving ample margin within the real-time requirements.
- **Robust Fallback:** If model checkpoints are missing, the agents fallback automatically to rule-based simulations to prevent system crashes.

Summary & Future Vision

GuardianAI represents a scalable, reliable driver assistance system designed for mass adoption:

- **Mass Adoption:** Low CPU footprint allows it to run on entry-level dashboards and standard smartphones.
- **Cloud Deployments:** Fully configured for Vercel deployment with dynamic WebSocket proxying.
- **Future V2X Integration:** Connects to surrounding cars over 5G to negotiate lane changes during emergency brake maneuvers.